Companion LXXV: The Fisher Kernel for Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

The Fisher kernel provides a principled method for combining information geometry with reproducing kernel Hilbert space (RKHS) methods. Applied to persistence diagrams, it yields feature maps that incorporate both the geometric structure of the underlying statistical manifold and the linear structure of kernel methods. Building on the Fisher information geometry introduced in Companion LXXIV, this work develops the Fisher kernel for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in the Fisher score and the resulting kernel. The Fisher kernel provides a powerful tool for classification, regression, and model discrimination in weak-lensing topology, bridging information geometry and RKHS-based topological data analysis.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak- lensing convergence fields. Previous Companions developed Hilbert-space embeddings, kernel PCA, maximum mean discrepancy (MMD), and Fisher information geometry as tools for analyzing and distinguishing the Relaxation-Driven Cyclic Cosmology (RDCC) from ΛCDM.

The Fisher kernel provides a natural synthesis of these approaches. It maps each persistence diagram to a feature vector given by the Fisher score, which measures the sensitivity of the log-likelihood to changes in cosmological parameters. The inner product of these score vectors defines a kernel that incorporates both information geometry and RKHS structure. This kernel is particularly well-suited for classification, regression, and hypothesis testing in cosmological topology.

2 Fisher Score and Fisher Kernel

Given a statistical model $p(D|\theta)$ for persistence diagrams D, the Fisher score is

$$U_\theta(D) = \partial_\theta \log p(D|\theta). \tag{1}$$

The Fisher information matrix is

$$I(\theta) = \mathbb{E}\left[U_\theta(D)U_\theta(D)^\top\right]. \tag{2}$$

The Fisher kernel is defined as

$$k_F(D_1, D_2) = U_\theta(D_1)^\top I(\theta)^{-1} U_\theta(D_2). \tag{3}$$

This kernel incorporates both the curvature of the statistical manifold and the local sensitivity of persistence diagrams to parameter changes.

3 RDCC Modification of Fisher Scores

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell)\left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}}\right)^\alpha\right]. \tag{4}$$

This modifies the distribution of persistence diagrams:

$$p_{\text{RDCC}}(D|\theta) = p(D|\theta)\left[1 - \chi_p \left(\frac{\ell}{\ell_{\text{rel}}}\right)^\alpha\right].$$

The Fisher score becomes

$$U_\theta^{\text{RDCC}}(D) = U_\theta(D)\left[1 - \chi_U \left(\frac{\ell}{\ell_{\text{rel}}}\right)^\alpha\right]. \tag{5}$$

4 The Fisher Kernel in RDCC

The RDCC Fisher kernel is

$$k_{F,\text{RDCC}}(D_1, D_2) = k_F(D_1, D_2)\left[1 - \chi_F \left(\frac{\ell}{\ell_{\text{rel}}}\right)^\alpha\right]. \tag{6}$$

RDCC induces:

- enhanced kernel similarity for large-scale topological modes,
- suppressed similarity for small-scale modes,
- a characteristic turnover near ℓ_{rel},
- modified Fisher curvature of the topological manifold,
- altered alignment of Fisher score vectors across redshift.

5 Applications to Cosmological Topology

The Fisher kernel enables:

- classification of RDCC vs. ΛCDM topological ensembles,
- regression of RDCC parameters from persistence diagrams,
- kernel PCA in Fisher space,
- Fisher-kernel MMD tests,
- information-geometric feature extraction for weak-lensing surveys.

These applications provide a powerful bridge between information geometry, kernel methods, and cosmological inference.

6 Conclusion

The Fisher kernel provides a principled method for combining information geometry with RKHS-based analysis of persistence diagrams. In the Relaxation-Driven Cyclic Cosmology, the relaxation mechanism modifies the Fisher score and the resulting kernel in a scale-dependent manner, producing distinctive signatures in weak- lensing topology. The present Companion establishes the foundation for future studies of Fisher-kernel methods, parameter inference, and information geometry in cosmological topology.

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